Parse

Volume 9

1 Background and Task

A spreadsheet formula comes in the following format:
`=[formula_name]([top left coordinates]:[bottom right coordinates])`.
Here are some examples:

1. `=SUM(A1:A1)`
2. `=PRODUCT(Z3:AA10)`
3. `=MAX(A1:A3)`

i.e., it always starts with an equal character '=', followed by a single word (made of only upper-case alphabetical letters), followed by an open parenthesis '(' , followed by a coordinate that specifies the top left corner, then a colon ':', followed by a coordinate that specifies the bottom right corner, then a close parenthesis ')'.
Note about the coordinates: The coordinate specifier comes in the form of (at most 3) alphabets for the column, followed by (at most 3) digits for the row. The alphabets follow the usual spreadsheet order: $A, B, C, \cdots, Z, AA, AB, AC \cdots, AZ, BA, BB, \cdots ..ZZ, AAA, \cdots$
In case you missed it: In this format, the columns are specified before the rows.
For example:

1. $A1$ specifies coordinate $0, 0$.
2. $B1$ specifies coordinate $0, 1$.
3. $Z1$ specifies coordinate $0, 25$.
4. $AA1$ specifies coordinate $0, 26$.
5. $AB1$ specifies coordinate $0, 27$.
6. $AB9$ specifies coordinate $8, 27$.
7. $AB10$ specifies coordinate $10, 27$.

Your Task: You need to read a single line from input. That line will specify a formula. Assume the input is always valid. Your job is to parse it into a struct for us. Details about the struct in the next section. (Don't worry we're not making Cells from the lecture.)

2 Code

You are given the following empty function(s) in `student.c` which you should implement. Only make changes within `student.c`.

```
FormulaInfo read_metadata_from_input(){

}
```

The linked list struct for your reference is given as:

```
typedef struct {
    size_t row;
    size_t col;
} Point;

typedef struct {
    char *formula_name;
    Point top_left;
    Point bottom_right;
} FormulaInfo;
```

The struct that you need to return needs to hold 3 things: The formula name, the top left corner (0-indexed), and the bottom right corner (0-indexed).
As an example, if `=SUM(A1:A1)` was the input string, the struct returned needs to be so that:

1. `formula_name` points to a heap-allocated (null-terminated) string whose contents are `"SUM"`
2. `top_left` should have value {0, 0}
3. `bottom_right` should have value {0, 0}

More examples shown below.

Implement all your code in `student.c`. To compile, run `make`, to test, run `make test`, and to manually run, do `./formula`, then input a formula and press ENTER.

Note: We will handle all printing to output for you. Just return the needed struct.
Note 2: You must allocate the string that goes in the struct. We will delete it for you.
Note 3: Yes, you have to handle input reading yourself. Read a single line of input. How you choose to do so is up to you. (We might give super long formula names).

3 Sample Input

(note that all the output will be on one line, we add a newline before "ending" to fit the PDF width)

Sample Input 1:

```
=SUM(A1:A1)
```

Expected Program Output 1:

```
Formula name: SUM, Starting from row, col: 0, 0,
ending on row, col: 0, 0
```

Sample Input 2:

```
=PRODUCT(A1:AA1)
```

Expected Program Output 2:

```
Formula name: PRODUCT, Starting from row, col: 0, 0,
ending on row, col: 0, 26
```

Sample Input 3:

```
=HELLO(A1:B2)
```

Expected Program Output 3:

```
Formula name: HELLO, Starting from row, col: 0, 0,
ending on row, col: 1, 1
```

Sample Input 4:

```
=HELLO(AA1:AA2)
```

Expected Program Output 4:

```
Formula name: HELLO, Starting from row, col: 0, 26,
ending on row, col: 1, 26
```

Sample Input 5:

```
=MAX(A1:AAA999)
```

Expected Program Output 5:

```
Formula name: MAX, Starting from row, col: 0, 0,
ending on row, col: 998, 702
```

Sample Input 6:

```
=SUPERCALIFRAGILISTICEXPIALIDOCIOUS(ZZZ1:ZZZ1)
```

Expected Program Output 6:

```
Formula name: SUPERCALIFRAGILISTICEXPIALIDOCIOUS,
Starting from row, col: 0, 18277,  ending on row, col: 0, 18277
```